

# Léa Mathieu - Embedded Software Engineer

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## PROFILE

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Recent graduate from ENSIMAG with a strong foundation in computer science, specializing in embedded systems. Passionate about reverse engineering, low-level development, and cybersecurity. Seeking an embedded software engineer position where I can leverage my skills in C/C++, as well as my experience with Zephyr and FreeRTOS, to contribute to innovative projects and grow within a dynamic company.

## EXPERIENCES

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### Embedded Software Apps Engineer

September 2025 – Present

*Texas Instruments*

*Oslo, Norway*

- Created a demo of a dual-RTOS solution by mixing Channel Sounding and Bluetooth Mesh capabilities to provide a one-chip solution for customers.
- Contributed to Zephyr drivers for efficient power management of a development board, significantly improving battery life and device performance.
- Provided technical support on Texas Instrument's engineer-to-engineer help forum, resolving complex Bluetooth Low Energy issues and improving customer satisfaction. Exhibited good communication skills and a customer-focused approach.

### Embedded Software Apps Engineer Intern

March 2025 – September 2025

*Texas Instruments*

*Oslo, Norway*

- Developed a Bluetooth Low Energy Wireless mouse using Zephyr and protocols such as SPI and UART, enhancing device connectivity and user interaction.
- Conducted direct video consultations with key customers to troubleshoot and resolve critical issues, ranging from MCUboot image validation to Bluetooth advertisement challenges. Showcased strong interpersonal skills and the ability to explain technical concepts clearly.

### Embedded Software Engineer Intern

June 2024 – Aug. 2024

*Allegro DVT*

*Grenoble, France*

- Facilitated the transition of a RISC-V micro-controller from FreeRTOS to Zephyr, optimizing system performance and resource management.
- Converted and adapted multiple drivers and libraries to ensure full compatibility with Zephyr, enhancing system stability and functionality.

## PROJECTS

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### x86 Operating System | C, x86, Linux

Jan. 2024 – June 2024

- Developed a fully functional x86 operating system from scratch using C, implementing core functionalities such as process management, memory management, and system calls. Led a team of 4 developers, coordinating tasks and ensuring project milestones were met in time.
- Designed and implemented a userspace and kernel separation to enhance system security and provide abstraction, which allowed for safer and more efficient process execution.

### Object Oriented Compiler | Java, ANTLR, Git, Docker, CI/CD, ARM

Dec. 2023 – Jan. 2024

- Served as the team leader for a group of 5 engineers in creating a comprehensive compiler for an object-oriented language akin to Java for ARM targets. Created collaterals to present the project to both a technical and non-technical audience.
- Utilized Gitlab runners to set up a continuous integration and delivery pipeline, automating the testing process to ensure code reliability and expedite development cycles.
- Implemented advanced language features such as inheritance, polymorphism, and exception handling, providing a robust tool for software development.

## TECHNICAL SKILLS

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**Languages:** C/C++, Python, x86, ARM, RISC-V, SQL (Postgres), JavaScript, Java

**Protocols:** UART, SPI, JTag, (Bluetooth Low Energy) BLE

**Framework:** Zephyr, FreeRTOS, MCUboot, React Native, Firebase, Stripe

**Developer Tools:** Git, Docker, Jenkins, Linux

## LANGUAGES

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**French:** Native

**English:** C1 Level – 985/990 TOEIC

**Spanish:** B1 Level

## EDUCATION

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### **ENSIMAG**

*Engineer School in Computer Science and Applied Mathematics*

Grenoble, France

*Sept. 2022 – June 2025*

### **Lycée Kléber**

*Intensive undergraduate program preparing for the nationwide competitive entrance exams*

Strasbourg, France

*Sept. 2020 – June 2022*